Hyperspectral Change Detection Based on Multiple Morphological Profiles

Zengfu Hou, Wei Li, *Senior Member, IEEE*, Lu Li, Ran Tao, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— With the increasing availability of multitemporal hyperspectral imagery, hyperspectral change detection under heterogeneous backgrounds is a challenging task. Due to the complexity of background features, traditional change detection algorithms in the spectral domain cannot effectively detect changed features. A novel method using multiple morphological profiles (MMPs) is proposed for hyperspectral change detection to make full use of spatial information. In the designed framework, first, the max-tree/min-tree strategy is applied to extract different attributes of multitemporal hyperspectral images (HSIs), i.e., area attribute and height attribute. Second, a spectral angle weighted-based local absolute distance (SALA) method is designed to reconstruct the discriminative spectral domain. Then, the absolute distance (AD) is adopted to extract changes in constructed feature domain. Finally, a change map is obtained by guided filtering. Experiments conducted on four real hyperspectral datasets demonstrate that the proposed detector achieves better detection performance.**

***Index Terms*— Change detection, guided filtering, hyperspectral image (HSI), morphological attribute profiles (APs).**

I. Introduction

As A key application of remote sensing, change detection can continuously observe and discover changed information of imaged scenes. Therefore, it is widely used in environmental monitoring, land change, urban expansion, disaster detection and evaluation, and military battlefield dynamic monitoring, and so on [1]–[3]. Different from high-resolution remote sensing images, hyperspectral images (HSIs) can provide fine spectral "diagnostic" information by collecting data into 3-D cubes with spatial and spectral dimensions, which offers potentials for accurate change detection and identification of change types [4]–[6].

According to different applications, various methods were proposed, and review articles have also classified and analyzed these methods from different aspects. Generally, HSI's change detection methods can be summarized into the following categories [7].

1) Algebra-based methods, which mainly include image difference, image ratio, image regression, absolute distance (AD) [8], change vector analysis [9], and so on. These methods judge the change of pixels by directly algebraic operation of multitemporal HSIs. Most algebra-based methods are based on the assumption that changes cause an obvious difference in pixel gray levels, and the pixel-by-pixel analysis is adopted. Therefore, the accuracy of radiometric and geometric correction results has a significant impact on detection accuracy.
2) Transformation-based methods, which mainly include the conventional principal component analysis (CPCA) and the temporal principal component analysis (TPCA) [10], multivariate alteration detection [11], the independent component analysis [12], and so on. These methods project the hyperspectral data into another feature space to mark the changed pixels or areas, which makes full use of spectral information but ignores continuous spectral signatures and similarity between adjacent pixels [13].
3) Classification-based methods [14]–[16]. It can be postclassification or direct classification method of multitemporal images. The postclassification method is a relatively intuitive change detection algorithm, in which images of different time series are classified separately, and then, classification results are compared and analyzed. Since these images of different time series are classified separately, the influence of the atmosphere, sensors, seasons, and other environmental factors on image acquisition can be eliminated. The direct classification method is to treat multitemporal images together and use a classifier method to find changed categories.
4) Other advanced methods, which mainly include unmixing-based [17]–[19], low rank and sparse representation-based [20], [21], deep learning-based methods [22]–[24], and so on.

These hyperspectral change detection methods are developed according to different purposes. Among these methods, deep learning-based methods [22]–[24] aim at generating a data-driven linear or nonlinear transformation to obtain advanced features of data for change detection. Although these methods have achieved superior change detection results, they are driven by data, so the scale of training database data and the accuracy of labels determine the performance

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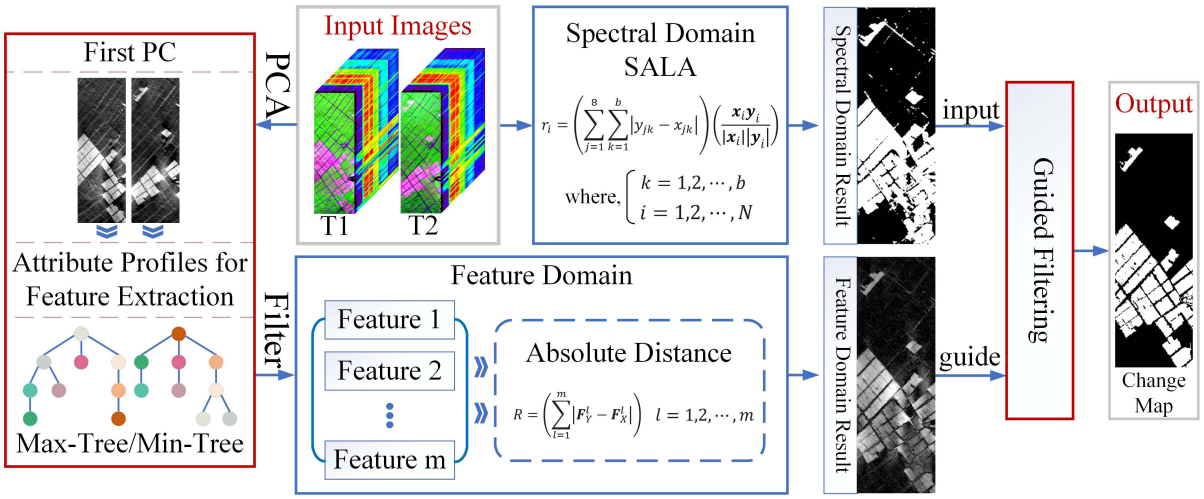


Fig. 1. Framework of the proposed dual-pipeline hyperspectral change detection framework using MMPs.

of these models. Therefore, these techniques are still under development.

Affected by various factors, such as seasons, terrain, weather conditions, and solar altitude angle, there are spectral variations in no-change pixels or areas changed information submerged in noise and background. Although these hyperspectral change detection algorithms have been widely used, there are still some problems, including: 1) lacking theoretical foundation and appropriate evaluation standards; 2) high sensitivity of data preprocessing; 3) poor versatility of detection method; 4) lacking multisource data integration analysis; and 5) low utilization of spatial information during detection. Therefore, how to increase the separability between changed pixels and background, and accurately identify the changed pixels from the background, has become a research hotspot in hyperspectral change detection.

With the development of graph theory in the field of remote sensing [25], it has been realized that the graph can reflect the topological structure between objects. Therefore, the tree theory is introduced into image processing to establish a dependency relationship between objects in the graph. Because the tree can well represent the relationship between pixels or areas, each node of the tree corresponds to a connected area in an image. Accurate and controllable feature extraction can be achieved by selecting a specific pruning strategy to filter the tree. Therefore, tree-based mathematical morphological algorithms are of great interest. Attribute profiles (APs) [26]–[28] have been applied in the field of remote sensing image processing, forming a unique morphological feature extraction method, where the component tree is selected to construct the topological structure of the image. The feature of the component tree is based on pixel value in the connected region. The root node represents the connected region with the smallest global pixel value, while the leaf node represents the connected region with the largest local pixel value. From root node to descendant node, a high-level relationship between connected regions in a local region is well-reflected.

Recently, morphological-based method [29]–[31] has shown some promising potential for infrared small target detection [32] applications, anomaly detection [27], and so on. However, to the best of our knowledge, morphological-based methods are not used in hyperspectral change detection. In this work, an effective hyperspectral change detection framework based on spectral and morphological features is designed, where an effective strategy using AP is presented. The max-tree/min-tree in AP is adopted to extract spatial size and contrast information of hyperspectral datasets in different time series. A spectral angle weighted-based local absolute distance (SALA) is also designed to further improve detection accuracy with the full use of the local neighborhood. Finally, a guided filtering method is adopted where the result of spectral analysis is guided by features to general a final change detection map.

The main contributions can be summarized as follows.

1) A dual-pipeline framework is proposed for hyperspectral change detection, where spectral and morphological features are fused by the guided filtering method.
2) In hyperspectral change detection, max-tree and min-tree in AP are used for the first time to extract morphological features of HSIs in different time series, where various morphological AP are utilized for change detection.
3) Moreover, in the process of the spectral domain, the SALA method is designed to further improve the effect by exploiting intrinsic local pixel relation.

The remainder of this article is organized as follows. In Section II, a detailed description of the proposed framework is presented. In Section III, four real datasets are utilized to verify the proposed method, and the experiment results and parameters are analyzed and discussed. The conclusion is drawn in Section IV.

II. Proposed Change Detection Framework

Fig. 1 illustrates the flowchart of the proposed morphological-based hyperspectral change detection framework, which consists of the following steps. First, the principal component analysis (PCA) [33] is adopted to obtain the first principal component of multitemporal HSIs. Then, a max-tree/min-tree method in AP is used to extract various features with pruning strategies. Furthermore, a designed SALA method is employed to enlarge the spectral

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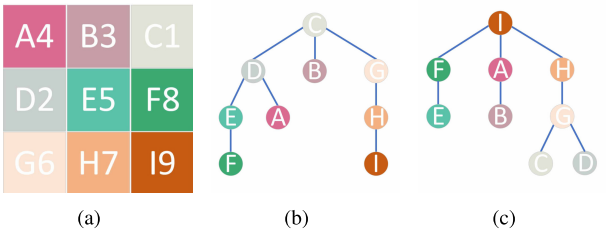


Fig. 2. Illustration of (b) max-tree and (c) min-tree. (a) Sample image.

difference between local pixels. Finally, a guided filter is adopted to generate a final detection map.

A. Multiple Morphological Profiles (MMPs)

Morphological representation can provide shape and size information related to objects and, therefore, has received extensive attention in HSI processing [26], [27], [32]. As a mathematical morphology algorithm, AP has been demonstrated its utility and rigor in mathematical description. In this work, a max-tree/min-tree strategy is first used to construct the tree structure of multitemporal images. For this purpose, max-tree/min-tree corresponding to the image should be constructed first, and then, adaptive attribute filter values should be selected to prune the constructed tree. Finally, by reconstructing the max-tree/min-tree into an image, features can be obtained. Morphological methods [28] are mainly used to process natural images. HSI contains hundreds of bands. If each band is processed by morphological filtering, it inevitably causes a very high-dimensional feature space [34]. Therefore, the PCA method is adopted to reduce the dimensionality of the original HSI, where the first principal component is selected for morphological feature extraction.

In Fig. 2(a), a sample image is shown where different colors represent various pixel values. The darker the color is, the greater the value is. The image-level sets can be represented by max-tree or min-tree shown in Fig. 2(b) or (c), where max-tree and min-tree of this sample are illustrated [32]. In the maximum tree, its value gradually increases from the root node to the leaf node, that is, the value of the leaf node is greater than that of the root node, and vice versa. Therefore, the structure of max-tree and min-tree is different. From Fig. 2(b), the value of root node of max-tree is 1, and leaf nodes are 8 and 9, respectively. That is, for the max-tree, the region of the root node (C) has the minimum pixel value, and leaf nodes (F, I) correspond to the maximum pixel values. However, the min-tree process is the opposite, as shown in Fig. 2(c), the region of the root node (I) has the maximum pixel value, and leaf nodes (C, D) correspond to the minimum pixel values.

B. Filter Process for Feature Extraction

After max-tree/min-tree being constructed, a pruning strategy [29], [30] is taken. The principle of attribute filtering is to keep branches of leaf nodes that meet requirements by setting threshold values and remove branches of leaf nodes that do not meet the requirements. If a node is filtered by a pruning strategy, then all its descendants are also filtered.

The attribute values are used to represent attributes of nodes in the tree, where attribute values of pixels or connected regions are calculated. In this article, five morphological attributes [32] are mainly used, including four increasing attributes, i.e., area attribute, height attribute, volume attribute, and the diagonal of bounding box attribute, and one nonincreasing attribute, i.e., standard deviation attribute. In these traditional attributes, the height is a contrast attribute, the area is a scale attribute, the volume is both a contrast attribute and a scale attribute, and the diagonal of the bounding box is a combination of shape and scale attribute. These attributes can be calculated as [32], [35]

$$\text{Area}(N) = \{\#p | p \in N\} \tag{1}$$

$$\text{Height}(N) = \text{Max}_{p\in N} f(p) - \text{Min}_{p\in N} f(p) \tag{2}$$

$$\text{Volume}(N) = \sum_{p\in N} (\text{Max}_{p\in N} g(p) - g(p)) \tag{3}$$

$$\text{Diag}(N) = \sqrt{(x_{p,\max} - x_{p,\min})^2 + (y_{p,\max} - y_{p,\min})^2} \tag{4}$$

where N denotes a connected region, p denotes a pixel in connected region, $f(p)$ is a function for the operation of obtaining the pixel value of p, $g = \pm f$ with the sign being determined according to the direction, and $x_{p,\max}$, $x_{p,\min}$, $y_{p,\max}$, and $y_{p,\min}$ are the extrema on the abscissa and ordinate in the connected region, respectively. $(x_{p,\max} - x_{p,\min})$ and $(y_{p,\max} - y_{p,\min})$ are the maximum height and width of the connected region, respectively.

Different from four morphological attributes mentioned above, standard deviation is a contrast attribute. The difference from other attributes is that standard deviation is a nonincreasing attribute. It can be calculated as

$$\text{Std}(N) = \sqrt{\frac{1}{\text{Area}(N)} \sum_{\forall p\in N} (f(p) - K_{\text{gray-level}}(\lambda))} \tag{5}$$

where Std is an abbreviation for standard deviation and $K_{\text{gray-level}}$ is the average intensity value of the pixel in the connected area, which can be given by

$$K_{\text{gray-level}}(\lambda) = \frac{1}{\text{Area}(N)} \sum_{\forall p\in N} f(p). \tag{6}$$

These morphological attributes are calculated differently, that is, the area attribute is to calculate the number of pixels in the connected region, the height attribute is to calculate the difference between pixels in the connected region and local pixel, the diagonal of the bounding box attribute is to calculate the variance of the connected region, and so on, so the mathematical meanings that they represented are also different and unique.

Pruning strategies are powerful tools, which can measure the persistence of regional extrema. Removal or preservation of the node is determined as follows. First, R_{N_i} is calculated according to the specific attribute, which is the corresponding result value. Second, η determines whether the node N_i should be removed or not, which is a threshold. If R_{N_i} is below η,

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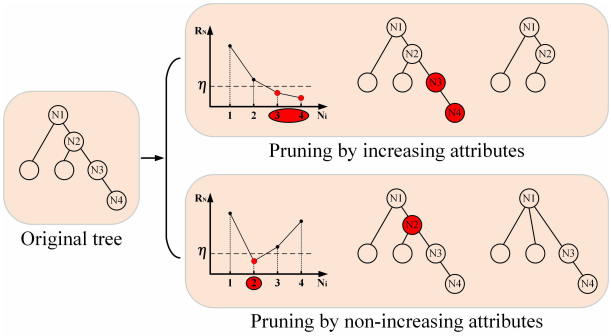


Fig. 3. Compare increasing attributes and nonincreasing attributes (red circles represent removed nodes, and others represent preserved nodes).

then the corresponding node is removed, and vice versa, which can be expressed as

$$N_i \quad \text{is} \begin{cases} \text{removed}, & \text{if } R_{N_i} < \eta \\ \text{preserved}, & \text{otherwise.} \end{cases} \tag{7}$$

As shown in Fig. 3, the pruning strategies by increasing and nonincreasing attributes are illustrated, where an original tree and two types results of different attributes for one local maximum are used for example. When root node N_1 is chosen, path to leaf node N_4 is $N_1-$to$-N_2-$to$-N_3-$to$-N_4$. In the coordinate system, the horizontal axis denotes node N_i (i is the ith node), and the vertical axis represents corresponding results R_{N_i}, where curves of increasing and nonincreasing attributes are also provided. For increasing attributes, the operation of tree filtering is straightforward when increasing attributes are employed. Since R_{N_3} and R_{N_4} are less than η, when N_3 is removed, N_4, which is descendant of N_3, is also removed. However, in nonincreasing attributes, only R_{N_2} is less than η, when N_2 is removed, and N_3 and N_4, which are the descendants of N_2, are preserved. Therefore, it can be concluded that all the descendants are removed when increasing attributes are used, but descendants may be preserved when the nonincreasing attributes are employed.

After pruning of max-tree/min-tree by attribute values, nodes on many branches are removed, and their corresponding connected regions are also covered by connected regions corresponding to the root node, which results in a change of connected regions corresponding to each node in the tree. A root node may be related to a connected region composed of multiple leaf nodes. When the tree is reconstructed into an image, useful feature information is retained, while useless feature information is deleted.

As shown in Fig. 4, the process of morphological max-tree is illustrated, which mainly includes max-tree construction, filtering, and image reconstruction. The two gray images that are the first principal component of the multitemporal images are used as initial images to construct the max-tree. Then, the structure of the max-tree is pruned with different morphological attributes' profile thresholds. After pruning, images represented by max-tree should be reconstructed into new feature images, in which information is preserved or deleted. Morphological features are stacked for processing.

C. Spectral Angle Weighted-Based Local Absolute Distance

After performing high-precision geometric registration on two images, AD is adopted as the basic change detection algorithm. The advantage of AD is simple and intuitive, and the result is easy to be interpreted, which can be expressed as

$$R_{\text{AD}} = \sum_{i=1}^{b} \left| \mathbf{B}_Y^i - \mathbf{B}_X^i \right|, \quad i = 1, 2, \ldots, N \tag{8}$$

where b is the band number, and $\mathbf{B}_X^i$ and $\mathbf{B}_Y^i$ represent the ith band at times T_1 and T_2, respectively.

The hypothesis of AD is that changed pixels or region may cause obvious change in pixel values, where the analysis of pixel by pixel is adopted. Therefore, radiometric correction and geometric correction have significant effect on detection result. Local background pixels have great influence on testing pixel, especially eight neighboring pixels around the testing pixel, which are very similar spectrally and spatially. In order to improve the robustness of AD algorithm, SALA is proposed, which can be expressed as

$$r_i = \left(\sum_{j=1}^{8} \sum_{k=1}^{b} |y_{jk} - x_{jk}| \right) \left(\frac{\mathbf{x}_i \mathbf{y}_i}{|\mathbf{x}_i||\mathbf{y}_i|} \right) \times k = 1, 2, \ldots, b \quad i = 1, 2, \ldots, N \tag{9}$$

where b and N denote the number of bands and pixels, respectively, $\mathbf{x}_i$ and $\mathbf{y}_i$ represent testing pixel corresponding to the scene at time T_1 and T_2, respectively, and y_{jk} and x_{jk} are pixel values corresponding a single band in eight neighboring pixels.

After an image is processed by morphological filtering, feature maps with different attributes can be obtained, which are stacked together to form a feature domain. Different from the spectral domain of the original image, the spatial feature domain cannot be directly used for object recognition because spectral information details are lost with the decrease in feature dimensions. Therefore, the performance of spectral-based SALA in the spatial domain is limited. However, this morphological feature is robust to illumination and shadow and can be used to recognize the change of background objects by the magnitude of the multidimensional difference feature map. In the proposed dual-pipeline framework, the SALA method is employed for the spectral domain, and the AD method is employed reformulated for the feature domain as

$$R = \sum_{l=1}^{m} \left| \mathbf{F}_Y^l - \mathbf{F}_X^l \right|, \quad l = 1, 2, \ldots, m \tag{10}$$

where m is the number of feature maps, and $\mathbf{F}_X^l$ and $\mathbf{F}_Y^l$ represent the lth feature at times T_1 and T_2, respectively.

D. Rectification With Guided Filtering

As effective edge-preserving filtering, guided filtering has been exploited in the field of image fusion, classification, feature enhancement, and so on. In the spectral domain, the SALA method integrates the change information on each band into a single band through weighted combination. However, this process may also cause amplification and accumulation of detection errors.

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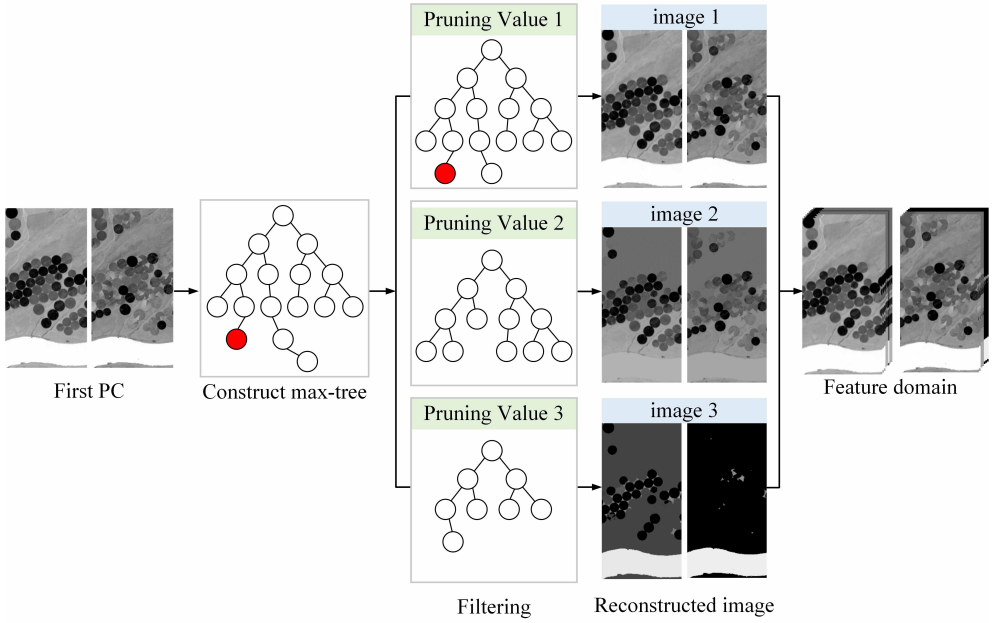


Fig. 4. Illustration of the pruning strategy using max-tree.

In this article, guided filtering is used to rectify the spectral domain detection map, where the spatial feature domain detection map is used as a guide map. The gray difference caused by acquisition conditions can be effectively reduced by using a morphological feature domain detection map to guide the spectral domain detection map. The guided filtering model [36] assumes that the output image q is a linear transformation of the guidance image g in a local window Ω_j centered at pixel j

$$q_i = a_j g_i + b_j, \quad \forall i \in \Omega_j \tag{11}$$

where a_j and b_j are the linear coefficients, Ω_j is a window of size $(2r+1) \times (2r+1)$, and i represents each pixel location in window Ω_j. In order to obtain coefficients a_j and b_j, it is transformed into an optimization problem and can be calculated by minimizing the difference between the input image p and the output image q in window Ω_j

$$E_{a_j,b_j} = \sum_{i \in \Omega_j} \left((a_j g_i + b_j - p_i)^2 + \varepsilon a_j^2 \right) \tag{12}$$

where ε is a regularization parameter. Since the window operation is adopted, the final guided filtering result can be obtained by taking the average value in the window

$$q_i = \bar{a}_i g_i + \bar{b}_i \tag{13}$$

where $\bar{a}_i$ and $\bar{b}_i$ are the average coefficients of all windows overlapping i. It is worth emphasizing that the guided filtering operation has been integrated into MATLAB R2018a. The built-in imguidedfilter function is used to perform guided filtering operation. The window size is [5 5] by default, which provides overall satisfying performance.

III. Experimental Results and Discussion

To evaluate the performance of the proposed method, four bitemporal datasets, acquired by the Hyperion sensor or the AVIRIS sensor, are conducted to perform hyperspectral change detection. It is worth being emphasized that these datasets, which are widely used in the field of hyperspectral changes, are obtained after preprocessing including atmospheric correction and geometric and radiometric correction [37].

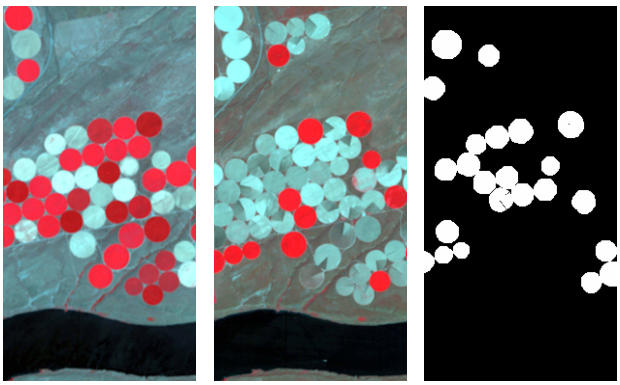
A. Datasets' Description

The first hyperspectral dataset is made of a pair of real bitemporal HSIs, collected from an irrigated agricultural field in Hermiston City in Umatilla County, Oregon, United States, which were acquired by the Hyperion sensor mounted onboard the Earth Observing-1 (EO-1) satellite May 1, 2004, and May 8, 2007, respectively [38], [39]. The Hermiston images have the size of 390 × 200 pixels and 242 spectral bands ranging from 0.4 to 2.5 μm. These images have a spectral resolution of 10 nm and a spatial resolution of 30 m. The main change on these datasets is the farmland land-cover change, including the transitions among crops, soil, water, and other land-cover types. The change occurred in the cropland is mainly due to vegetation water content affected by irrigation condition and the difference of crop growth situation [39]. The scene and the ground-truth map are shown in Fig. 5.

The second dataset contains two HSIs for change detection, which were collected for a river area in Jiangsu Province, China, in May 3, 2013, and December 31, 2013, respectively. The sensor used is also the Earth Observing-1 (EO-1) Hyperion, with a spectral range of 0.4–2.5 μm, a spectral resolution of 10 nm, a spatial resolution of 30 m, and a total of 242 spectral bands. The size of the image in the dataset is 463 × 241 pixels, and 198 bands are used. The main change type is the disappearance of substance in the river [22], [38]. The images and the ground-truth map are illustrated in Fig. 6.

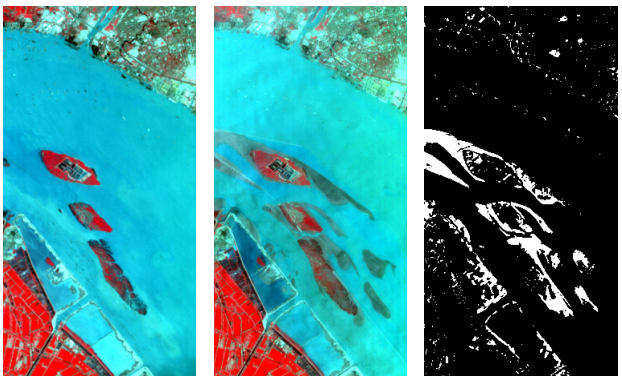
As shown in Fig. 7, the third dataset, including two HSIs, is the Santa Barbara scene, taken in 2013 and 2014,

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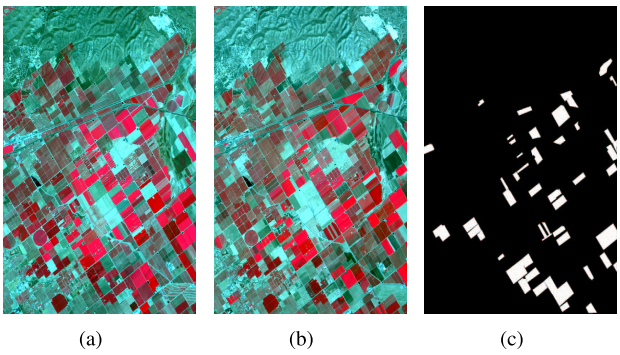
(a) (b) (c)

Fig. 5. Illustration of the Hermiston dataset. (a) Farmland on May 1, 2004. (b) Farmland on May 8, 2007. (c) Ground-truth change map.



(a) (b) (c)

Fig. 6. Illustration of the River dataset. (a) River scene on May 3, 2013. (b) River scene on December 31, 2013. (c) Ground-truth change map.

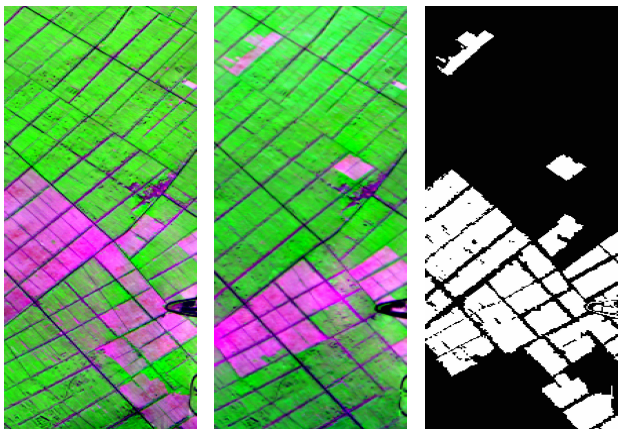


(a) (b) (c)

Fig. 7. Illustration of the Santa Barbara dataset. (a) Santa Barbara scene on 2013. (b) Santa Barbara scene on 2014. (c) Ground-truth change map.

respectively, with AVIRIS sensor over the Santa Barbara region in California, United States. Its spatial dimension is 984 $\times$ 740 pixels and includes 224 spectral bands [40]. A reference change map of the object-based areas was constructed by visual inspection. The detailed description can be found in [40]. It is worth emphasizing that the difference of change areas is not obvious, which may be caused by different channels selected for pseudocolor display.

The fourth dataset, including two hyperspectral bitemporal images, was obtained by EO-1 Hyperion sensor, which were collected over a wetland agricultural area in Yancheng city,



(a) (b) (c)

Fig. 8. Illustration of the Yancheng dataset. (a) Farmland on May 3, 2006. (b) Farmland on April 23, 2007. (c) Ground-truth change map.

Jiangsu Province, China, in May 3, 2006, and April 23, 2007, respectively [37]. Subset images with 420 $\times$ 140 pixels and 154 spectral bands were used after removing noisy and water-absorption bands. The images and the ground-truth map are illustrated in Fig. 8.

B. Parameters' Setting

In order to validate the proposed morphological-based method effectively, results derived by the proposed method are compared with those derived by five other methods, including AD [8], absolute average difference (AAD) [9], [41], subspace-based change detection (SCD) [42], local SCD (LSCD) [42], and adaptive SCD (ASCD) [42].

In the contrast methods, such as LSCD and ASCD, their performance is sensitive to varying window sizes. However, in the proposed max-tree/min-tree-based method, its performance is insensitive to pruning thresholds although suitable thresholds can indeed improve the accuracy of change detection. In the proposed morphological-based method, the parameter setting is listed in Table I, where Diag_box represents the diagonal of the bounding box.

For the proposed morphological-based method, after constructing the max-tree/mini-tree, a suitable threshold value should be considered to prune the shape of the constructed tree, which is the filtering operation in morphological processing. Among the five attributes, ten thresholds are set for each attribute, and the minimum threshold is set 10. The interval of different thresholds in the area attribute is 5, and the interval of other attributes is 3. It is worth noting that these parameters are set empirically. In current morphological-based algorithms, there is no effective automatic threshold selection strategy. Most of the thresholds are scene-specific. In this article, because the threshold interval of the area attribute is too small, the difference between various features is blurred, so here the interval of different thresholds is set as 5, and the interval of the other four parameters is set as 3.

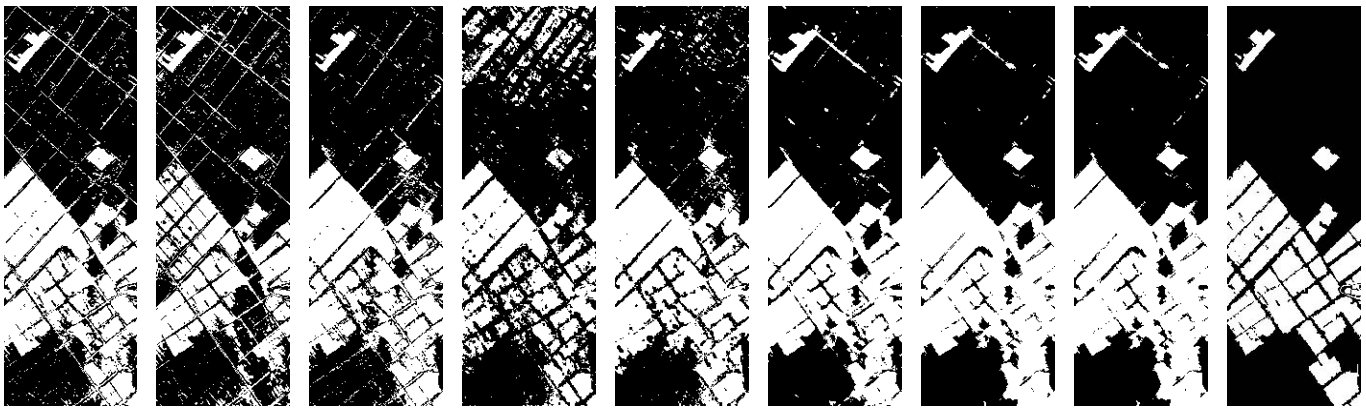
For the window size, in LSCD and ASCD, by changing the window size from 3 to 15, detection performance under different window sizes is collected, and then, the optimal

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TABLE I

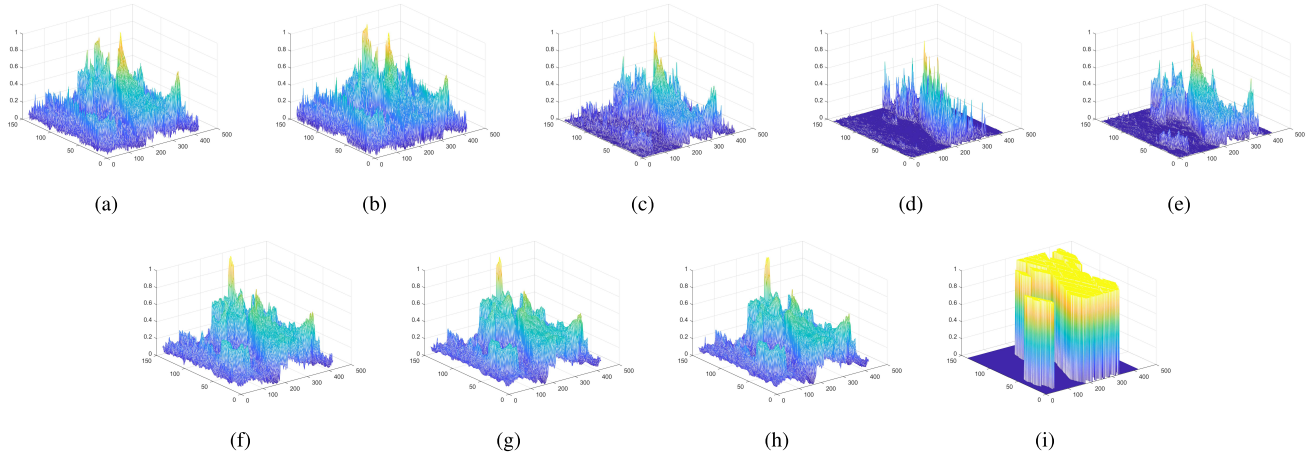
THRESHOLDS SETTING UNDER DIFFERENT ATTRIBUTES IN MAX-TREE/MIN-TREE

Attributes	Area	Height	Volume	Diag_box	Std
Thresholds	10,15,20,25,30, 35,40,45,50,55	10,13,16,19,22, 25,28,31,34,37	10,13,16,19,22, 25,28,31,34,37	10,13,16,19,22, 25,28,31,34,37	10,13,16,19,22, 25,28,31,34,37



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 9. Illustration of binary detection results when the false alarm rate reaches 0.15 in the Yancheng dataset. CDR. (a) AD: 0.898. (b) AAD: 0.697. (c) SCD: 0.892. (d) LSCD: 0.781. (e) ASCD: 0.910. (f) SALA: 0.919. (g) MaxtreeCD: 0.941. (h) MintreeCD: 0.940. (i) Ground truth.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. Illustration of 3-D detection results in the Yancheng dataset. (a) AD. (b) AAD. (c) SCD. (d) LSCD. (e) ASCD. (f) SALA. (g) MaxtreeCD. (h) MintreeCD. (i) Ground truth.

TABLE II

OPTIMAL WINDOW SIZE OF VARIOUS METHODS IN DIFFERENT DATASETS

Methods	LSCD	ASCD
Hermiston	3	13
River	3	3
Santa Barbara	5	5
Yancheng	3	3

parameter is selected [43]. The optimal window sizes of various methods in different datasets are listed in Table II.

C. Results and Discussion

Taking the Yancheng dataset as an example, detection results of various algorithms are displayed in Figs. 9 and 10. In Fig. 9, binary detection results are displayed when the false alarm rate reaches 0.15 in the Yancheng dataset. It can be concluded that SALA, maxtreeCD and mintreeCD can achieve better detection results while suppressing unchanged pixels. Compared to Fig. 9(a) and (b), it can be found that the proposed SALA has stronger suppression ability to unchanged pixels and better discrimination ability to changed pixels, which also shows that the proposed SALA is more effective than AD. Similarly, from the specific change detection rate (CDR) of various algorithms, as listed in Table III, the same conclusion can be drawn. When the false alarm rate reaches 0.15, SALA, maxtreeCD, and mintreeCD have better detection than others. In Fig. 10, 3-D detection results of various algorithms are more intuitive. In Fig. 10(c)–(e), it is clearly observed that SCD, LSCD, and ASCD suppress unchanged and changed pixels simultaneously. Therefore, the height of the detected changed pixels in 3-D results is

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TABLE III
CHANGED DETECTION RATE ON THE YANCHENG DATASET WHEN FALSE ALARM RATE REACHES 0.15

Methods	AD	AAD	SCD	LSCD	ASCD	SALA	maxtreeCD	mintreeCD
CDR	0.898	0.697	0.892	0.781	0.910	0.919	0.941	0.940

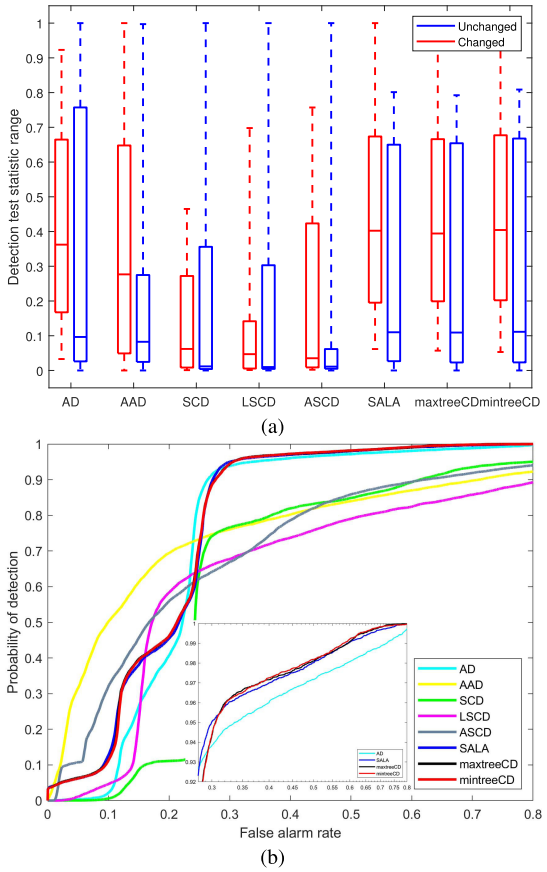


Fig. 11. ROC curves of different methods in the Hermiston dataset. (a) Statistical separability analysis. (b) ROC curves.

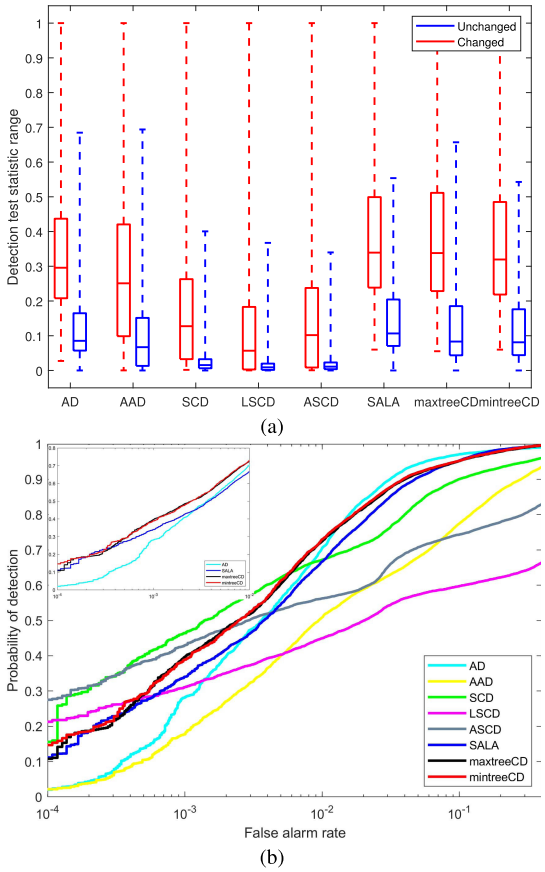


Fig. 12. ROC curves of different methods in the River dataset. (a) Statistical separability analysis. (b) ROC curves.

lower than that of other algorithms. Fig. 10(f)–(h) shows better detection than other algorithms but still cannot offer obvious distinction.

To quantitatively compare the performance of various algorithms, the receiver operating characteristic curve (ROC) and area under curve (AUC) metrics are utilized [44]. Meanwhile, the statistical separability analysis, also known as boxplot [45], is employed for performance assessment. It is mainly used to reflect the distribution characteristics of original data and can also be used to compare distribution characteristics of multiple groups of data. For the statistical separability analysis, as shown in Figs. 11(a)–14(a), red and blue boxes represent the range of changed and unchanged pixels, respectively. The interval between red and blue boxes represents the separability between changed and unchanged pixels. The height of the box represents suppression degree to changed and unchanged pixels. Generally, the lower the blue box height is, the stronger the unchanged pixels suppression is, which is also conducive to the separation of changed and unchanged pixels. The horizontal line in the middle of red or blue boxes represents the median of changed or unchanged pixels and has a positive correlation with the mean to some extent.

For the Hermiston dataset, the separability between changed and unchanged pixels, and the output value range of different detection methods are displayed in Fig. 11(a). In ASCD, the height of the blue box is significantly lower than that of the red box, which shows that the algorithm can produce good suppression on unchanged pixels. Similarly, the AAD also suppresses unchanged pixels. However, other algorithms, such as AD, SCD, LSCD, SALA, maxtreeCD, and mintreeCD, cannot effectively suppress unchanged pixels, but this does not mean that these algorithms cannot effectively extract changed pixels. From the horizontal line in the middle of the red box or blue box, it can be found that, compared with others, the difference between changed and unchanged median intervals in SALA, maxtreeCD, and mintreeCD is obvious, which also shows that the proposed algorithms can separate the changed and unchanged pixels to some extent. ROC curves of various detection algorithms are displayed in Fig. 11(b). It is easy to find that, when the false alarm rate reaches

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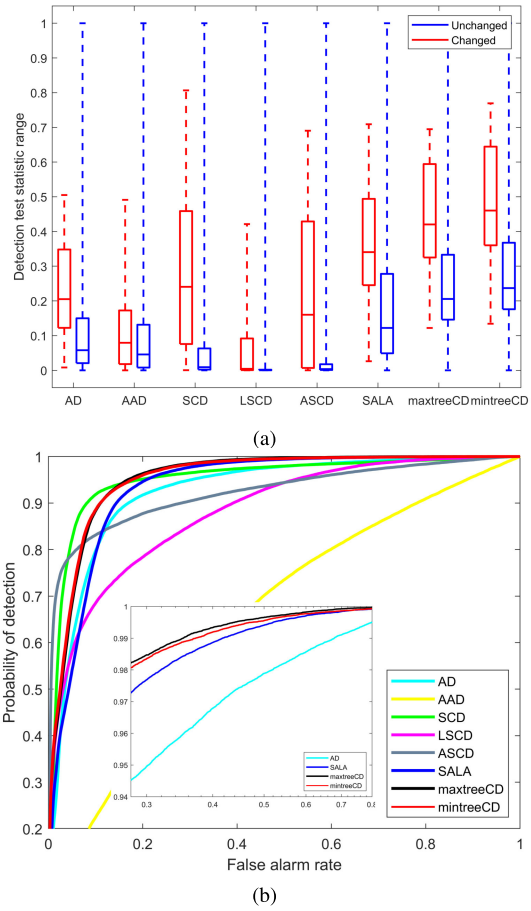


Fig. 13. ROC curves of different methods in the Santa Barbara dataset. (a) Statistical separability analysis. (b) ROC curves.

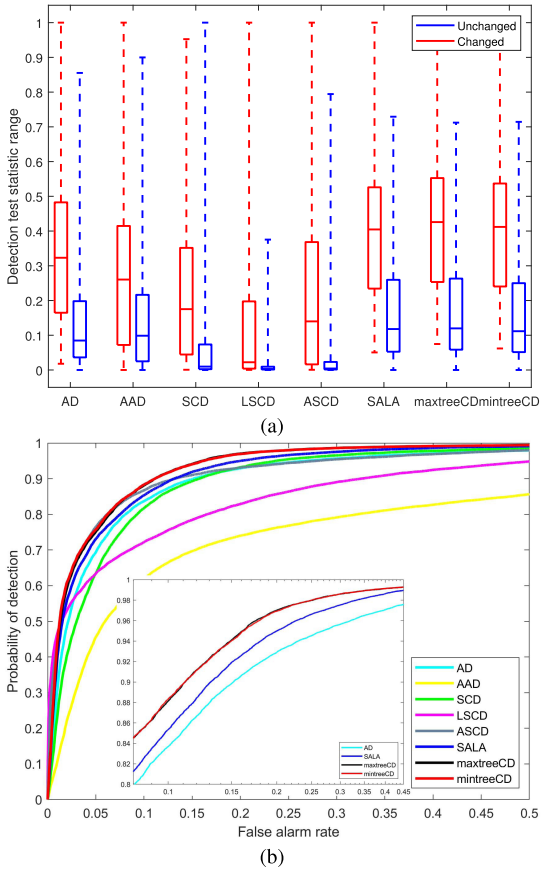


Fig. 14. ROC curves of different methods in the Yancheng dataset. (a) Statistical separability analysis. (b) ROC curves.

about 0.28, blue, black, and red ROC curves representing the proposed SALA, maxtreeCD, and mintreeCD, respectively, are obviously on the upper left corner, which reflects that the performances of SALA, maxtreeCD, and mintreeCD are better than other methods. The cyan curve is closest to the red and black curves, which also reflects that the performance of AD is closest to the performance of SALA, maxtreeCD, and mintreeCD.

For the River dataset, from the statistical separability analysis, as shown in Fig. 12(a), it can be easily observed that interval between red and blue boxes of the proposed SALA, maxtreeCD, or mintreeCD method are obviously larger than other methods, which indicates that the proposed method can separate changed pixels from unchanged pixels more effectively. In SCD, LSCD, and ASCD, the height of the blue box is significantly lower than others, which means that SCD, LSCD, and ASCD have the better changed pixels suppression compared with other methods. However, the performance of SCD, LSCD, and ASCD is poor because these methods have the poor ability of suppression and separation of unchanged pixels. In Fig. 12(b), compared with other detection methods, the curves of maxtreeCD, mintreeCD, and AD are slightly on the upper left corner, which shows that these three algorithms can outperform. However, from the enlarged picture, it can be confirmed that the effectiveness of the proposed two strategies is superior to AD. Similarly, for the Santa Barbara dataset, as shown in Fig. 13(a), the same conclusion can be drawn that LSCD and ASCD can better remove unchanged pixels. The SALA, maxtreeCD, and mintreeCD algorithms can separate changed and unchanged pixels to some extent. In Fig. 13(b), the ROC curves of the proposed maxtreeCD and mintreeCD are closer to the upper left corner, which also means that detection performances of the proposed maxtreeCD and mintreeCD are better. In the fourth dataset, which is the Yancheng dataset, the statistical separability analysis, and the ROC curve are shown in Fig. 14, where the same conclusion can be drawn that the proposed maxtreeCD and mintreeCD present optimal or suboptimal detection performances.

Table IV lists AUC values. In the spectral domain, the proposed SALA method shows better detection effects than AD in three datasets. The detection results of the proposed maxtreeCD and mintreeCD are better than other methods for these four datasets; especially for the Yancheng dataset, compared with the AD method, the improvement of mintreeCD is approximately 2.2%. Through the analysis of the ROC curve and AUC, it can be further confirmed that the proposed maxtreeCD and mintreeCD are more effective than SALA, AD, and other traditional change detection algorithms, which shows that combining morphological features can effectively improve the change detection ability.

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TABLE IV
ACCURACY COMPARISON OF BINARY CD ON FOUR DIFFERENT DATASETS (AUC VALUES %)

Methods	AD	AAD	SCD	LSCD	ASCD	SALA	maxtreeCD	mintreeCD
Hermiston	78.582	77.828	68.348	68.236	74.111	80.445	80.575	80.536
River	98.002	91.582	95.038	74.182	85.530	97.950	98.065	98.141
Santa Barbara	92.375	65.826	95.131	88.148	95.238	93.405	95.240	95.266
Yancheng	93.696	81.365	93.443	89.878	94.545	95.004	95.852	95.913

TABLE V
EXECUTION TIME OF DIFFERENT METHODS USING FOUR EXPERIMENTAL DATASETS (UNIT: SECONDS)

Methods	AD	AAD	SCD	LSCD	ASCD	SALA	maxtreeCD	mintreeCD
Hermiston	0.048	0.044	0.277	21.388	28.097	4.120	11.556	12.588
River	0.053	0.052	0.338	27.333	3.345	5.031	30.850	38.754
Santa Barbara	0.412	0.361	2.493	569.538	69.505	38.495	1364.040	1346.664
Yancheng	0.022	0.020	0.147	12.495	1.535	2.116	16.757	11.328

Table V provides the computational cost of various detection methods. All the experiments are conducted on Windows 10 with a 64-b operating system. The processor of the system is an Intel Core i7-8700 CPU with 3.20 GHz with 16 GB of main memory. The code was implemented in VISUAL STUDIO 2017 and MATLAB R2018a. The feature extraction in max-tree/min-tree was conducted in VISUAL STUDIO 2017 using C++ language, and the rest detection was performed in MATLAB R2018a, so the execution time is the sum of the time consumed by C++ and MATLAB. As shown in Table V, compared with other traditional change detection algorithms, the computational cost of maxtreeCD and mintreeCD methods is still high, especially for the Santa Barbara dataset. The reason is that the morphological-based feature extraction process is time-consuming.

IV. CONCLUSION

In this article, a novel space-spectral feature dual-pipeline framework was designed for hyperspectral change detection. The changed features in the morphological spatial domain and the spectral domain were effectively fused together through guided filtering. To the best of our knowledge, the morphological APs were the first applied to hyperspectral change detection, where the max-tree and min-tree strategies were used to extract various types of pixels or regions whose brightness varies greatly in the bitemporal datasets. Furthermore, a SALA method was designed to further enlarge the difference between local pixels and central pixels in the spectral domain. Experimental results on four real datasets demonstrated that the proposed strategy has better detection performance than other methods. However, the computational cost of the proposed method is high, which will be the focus of our future work.

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